

Customer Churn Prediction System

End-to-End ML System with CRISP-DM & MLOps (MLflow)

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System Architecture

- Data ingestion pipeline
- Feature engineering pipeline
- ML model training (MLflow tracking)
- Model registry
- API inference service

Churn Prediction Workflow

- Customer data input
- Feature transformation
- Model inference
- Churn probability output
- Business action (retention campaign)

- REST API (FastAPI / Flask)
- Docker containerization
- Model served from MLflow registry

Production Monitoring

- API latency and error tracking
- Data drift detection
- Prediction distribution monitoring
- Model performance degradation alerts

End-to-End MLOps Pipeline

Data → Features → Training → MLflow → Deployment → Monitoring → Retraining

Challenges

- Class imbalance in churn data
- Feature leakage risks
- Model interpretability vs performance tradeoff

Future Improvements

- Real-time churn scoring
- Automated retraining pipeline
- A/B testing retention strategies
- Improved feature store integration

Conclusion

- Built end-to-end churn prediction system
- Achieved strong predictive performance ($F1 = XX\%$)
- Implemented full MLOps lifecycle with MLflow
- System ready for production decision support

Thank You